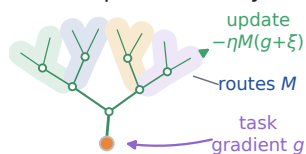


A

Credit-operator utility

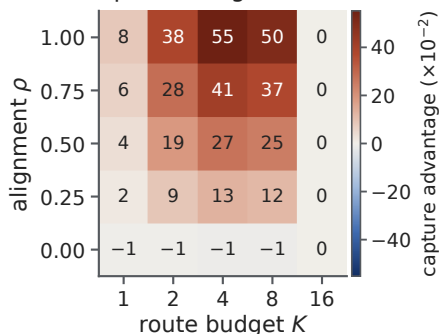


$$\text{guarantee} \propto \frac{[g^T M g]^2}{\|M g\|^2 + \text{tr}(M \Sigma M^T)}$$

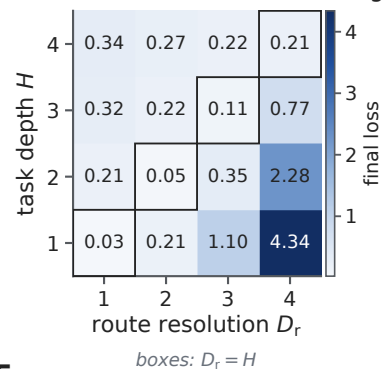
signal² / (gain + noise)

B

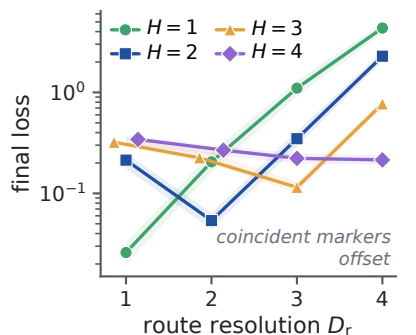
Spectral alignment

**C**

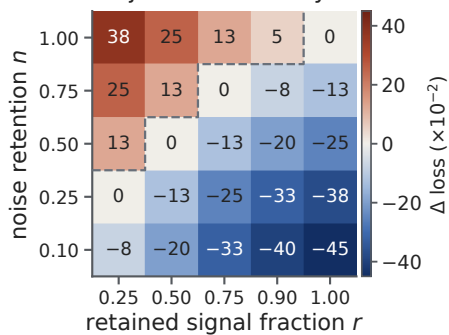
Route-resolution matching

**D**

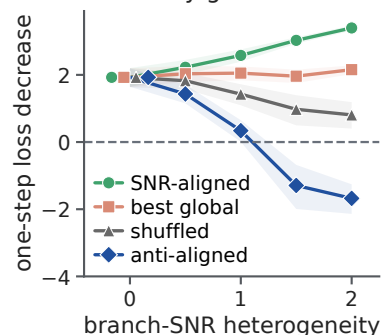
Route-resolution crossover

**E**

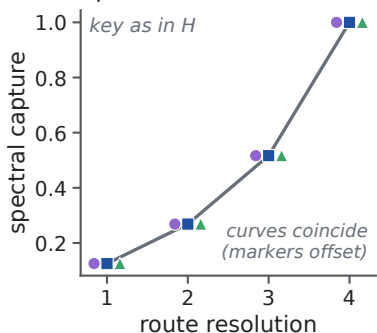
Projection boundary

**F**

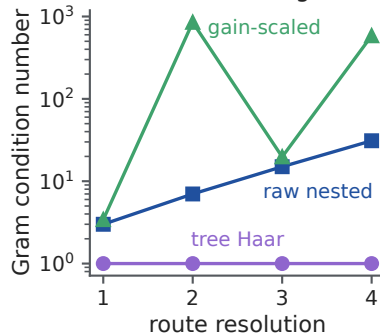
Reliability gains

**G**

Span invariance

**H**

Route conditioning

**I**

Predictive utility

